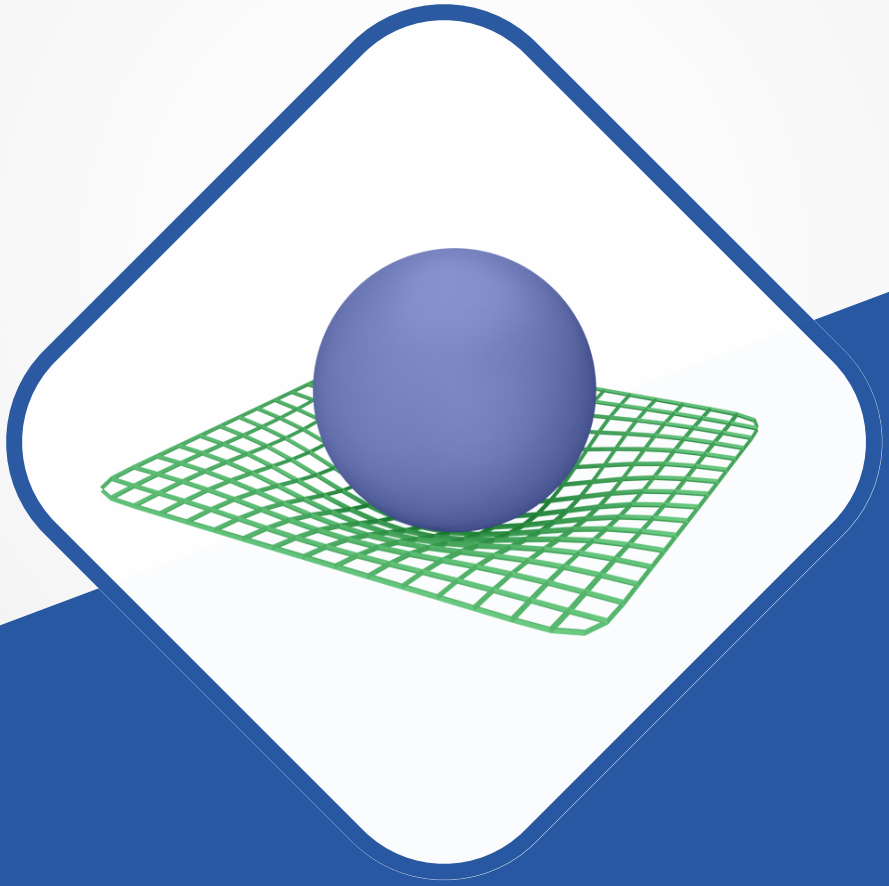
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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JEE (MAIN+ADVANCED)

Physics

ENTHUSIAST COURSE



STUDY MATERIAL

Gravitation

English Medium

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GRAVITATION

KEY CONCEPT

1. NEWTON'S LAW OF GRAVITATION

According to Newton's law of gravitation every particle in the universe attracts every other particle with a force directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of this force is along the line joining the particle. Thus, the magnitude of the gravitational force F that two particles m_1 and m_2 separated by a distance r exert on each other is

$$F = G\frac{m_1 m_2}{r^2}$$

Here G, called *gravitational constant*, is a universal constant having the value = 6.67×10^{-11} N-m^2/kg^2 and dimensions [$M^{-1}L^3T^{-2}$].

Regarding gravitational force it is worth noting that :

(1) It is always attractive (while electric and magnetic force can be attractive or repulsive).

(2) It is independent of the medium between the particles (while electric and magnetic forces depend on the nature of the medium between the particles).

(3) It holds good over a wide range of distances. It is found true from interplanetary distances to interatomic distances.

(4) It is an action-reaction pair, i.e., the force with which one body (say earth) attracts the second body (say moon) is equal to the force with which moon attracts the earth. (However, accelerations will not be equal)

(5) It is a two-body interaction, i.e., gravitational force between two particles is independent of the presence or absence of other particles; so the principle of superposition is valid, i.e., force on a particle due to number of particles is the resultant of forces due to individual particles, i.e.,

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + ...$$

(6) It is a conservative force, i.e., work done by it is path independent or work done in moving a particle round a closed path under the action of gravitational force is zero.

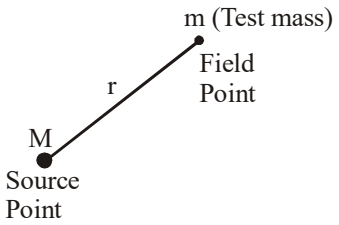
(7) It expresses the force between point masses. In applying it to the case of extended bodies of finite size care should be taken in assuming that the whole mass of a body can be regarded as concentrated at its centre of mass.

Gravitational Intensity $\vec{E}_G$

In a gravitational field, at a point, the gravitational intensity $\vec{E}_G$ is defined as force experienced per unit mass acting on a test mass supposed to be placed at that point.

$$\vec{E}_G = \vec{F}/m$$

It is a vector having dimensions [LT^{-2}] and unit N/kg.



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Gravitational Potential V

At a point in a gravitational field potential V is defined as negative of the work done per unit mass in shifting a rest mass from some reference point (usually at infinity) to the given point, i.e.,

$V = (-W/m)$

(1) It is a scalar having dimensions $[L^2T^{-2}]$ and unit J/kg.

(2) As by definition of work $W = \int \vec{F} \cdot \vec{dr}$

So $V = -\int \frac{\vec{F} \cdot \vec{dr}}{m} = -\int \vec{E}_G \cdot \vec{dr}$ [as $\frac{\vec{F}}{m} = \vec{E}_G$]

i.e., $dV = -E_G\, dr$ or $E_G = -(dV/dr)$

For Formula conversion :

Electrostatics	**Gravitation**
Q	M
K	G
$\in_0$	$\frac{1}{4\pi G}$

Point Charge

$E = \frac{kQ}{r^2}$

$V = \frac{kQ}{r}$

Point Mass

$E_G = \frac{GM}{r^2}$

$V_G = \frac{-GM}{r}$

Uniform charged ring

$E = \frac{kQx}{(r^2 + x^2)^{3/2}}$ on axis

E is max. when $x = \frac{r}{\sqrt{2}}$

$V = \frac{kQ}{\sqrt{r^2 + x^2}}$ on axis , $\frac{kQ}{r}$ at center

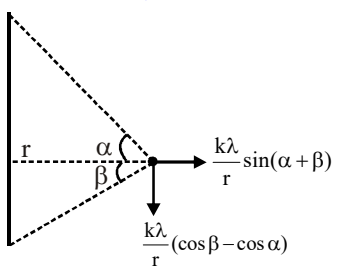
Ring of uniform mass distribution

$E_G = \frac{GMx}{(r^2 + x^2)^{3/2}}$ on axis

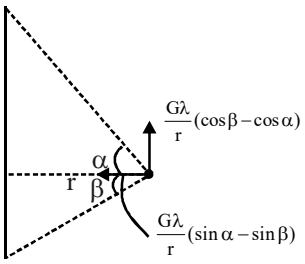
E_G is max. when $x = \frac{r}{\sqrt{2}}$

$V_G = \frac{-GM}{\sqrt{r^2 + x^2}}$ on axis, $\frac{-GM}{r}$ at center

Uniform linear charge



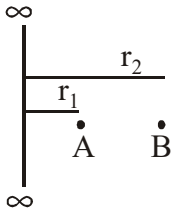
Uniform linear mass



(λ=mass per unit length)

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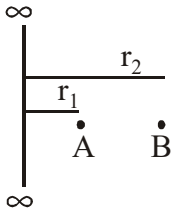
Infinite Linear charge



$$E = \frac{2K\lambda}{r}$$

$$V_B - V_A = -2K\lambda \ln\frac{r_2}{r_1}$$

Infinite linear mass



$$E_G = \frac{2G\lambda}{r}$$

$$V_B - V_A = 2G\lambda \ln\left(\frac{r_2}{r_1}\right)$$

Infinite Sheet of charge

$$E = \frac{\sigma}{2\varepsilon_0}$$

Infinite Sheet of mass

$$E_G = \frac{\sigma}{2} \times 4\pi G = 2\pi G\sigma$$

(σ = mass per unit area)

Uniformly charged hollow shpere

Charge Q, radius R

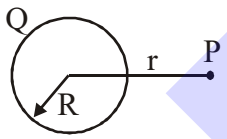
distance of field point from center r

Hollow sphere of uniform mass

Mass M , radius R

distance of field point from center r

Case I r > R



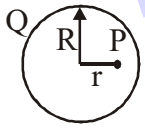
$$E = \frac{kQ}{r^2}$$

$$V = \frac{kQ}{r}$$

$$E_G = \frac{GM}{r^2}$$

$$V_G = -\frac{GM}{r}$$

Case II r < R



$$E = 0$$

$$V = \frac{kQ}{r}$$

$$E_G = 0$$

$$V_G = -\frac{GM}{R}$$

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Uniformly charged solid sphere mass M, radius R	**Uniformly solid sphere** mass M, radius R
$E = \frac{kQ}{r^2}$, $r > R$	$E_a = \frac{GM}{r^2}$, $r > R$
$\frac{kQr}{R^3}$, $r < R$	$\frac{GM}{R^3}r$, $r < R$
$V = \frac{kQ}{r}$, $r > R$	$V_a = -\frac{GM}{r}$, $r > R$
$\frac{kQ}{2R^3}\left(3R^2 - r^2\right)$, $r > R$	$\frac{-GM}{2R^3}\left(3R^2 - r^2\right)$, $r > R$

2. ACCELERATION DUE TO GRAVITY 'g'

Force of gravity (of reference body) in accordance with Newton's II law (i.e., F = ma) will produce an acceleration in it. *The acceleration produced in a body by the force of gravity of reference body (usually earth) is called acceleration due to gravity* and is represented by 'g'.

(1) It is a vector having dimensions $[LT^{-2}]$, SI unit m/s^2.
Its value near the surface of earth is 9.8 m/s^2.

(2) As by definition $\vec{g} = \left(\vec{F}_g / m\right)$ and also $\vec{E}_G = \left(\vec{F}_G / m\right)$ so $\vec{g} = \vec{E}_G$, i.e., acceleration due to gravity and gravitational intensity E_G at a point are equal in magnitude and direction.

(3) If the reference body has mass M and radius R, the force on a body of mass m at the surface of reference body by Newton's law of gravitation will be

$$F_G = \left(\frac{GMm}{R^2}\right) \quad \text{... (i)}$$

and if 'g' is acceleration due to gravity of reference body on its surface, by Newton's II law,

$$F_g = mg \quad \text{... (ii)}$$

As Equation (i) and (ii) represent the same,

$$g = \frac{GM}{R^2} \quad \text{... (iii)}$$

3. VARIATION OF 'g'

(i) *Due to altitude*: Acceleration due to gravity at a height h above the surface of earth is given by:

$$g_h = \frac{GM}{(R+h)^2} = g\left[1 - \frac{2h}{R}\right] \quad \text{(for } h << R\text{)}$$

(ii) *Due to depth:* Acceleration due to gravity at a depth h below the surface of earth is given by

$$g_h' = g\left[1 - \frac{h}{R}\right] \quad \text{(for all depths)}$$

At h = R (i.e. at the centre of earth): $g_h' = 0$

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(iii) *Due to rotation of earth:* Acceleration due to gravity at latitude λ is given by:

$g_\lambda = g - R\omega^2\cos^2\lambda$ where ω = angular velocity of the earth

(a) ***At poles*** : $g_p = g - R\omega^2\cos^2(\pi/2) = g = g_{max}$

(b) ***At equator*** : $g_{eq} = g - R\omega^2\cos^2 0 = g - R\omega^2 = g_{min}$

(iv) *Due to non-spherical shape of earth:* Due to the shape of the earth, g is maximum at poles and minimum at equator.

4. ESCAPE VELOCITY

(i) It is minimum speed with which a body must be projected away from the surface of the earth so that it may never return to the earth.

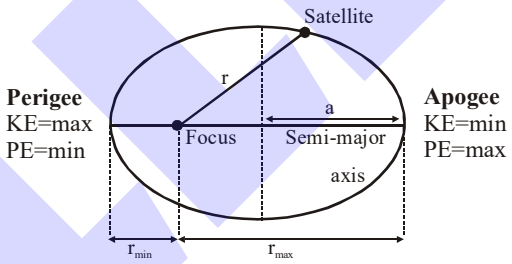
(ii) Escape velocity of a body from the surface of earth is given by : $v_{es} = \sqrt{2gR} = \sqrt{(2GM/R)}$

5. KEPLER'S LAWS:

(i) All planets move around the sun in **elliptical orbits**, with the sun being at rest at one focus of the orbit.

(ii) The position vector from the sun to the planet sweeps out equal area in equal time, i.e., areal velocity of a planet around the sun always remains constant. This gives that the **angular momentum remain constant**.

(iii) The square of the time period of a planet around the sun is proportional to the cube of the semi-major axis of the ellipse or mean distance of the planet from the sun, i.e. $T^2 \propto a^3$ where a is the semi-major axis of the ellipse.



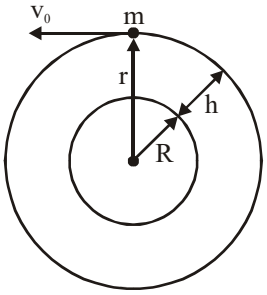
6. SATELLITE MOTION

(a) Orbital Velocity v_0 : Let a satellite of mass m be moving in an orbit of radius r with speed v_0 as shown in figure, the centripetal force required to the motion is $\left(mv_0^2/r\right)$ while the gravitational force is $\left(GMm/r^2\right)$. As gravitation provides centripetal force,

$$\frac{mv_0^2}{r} = \frac{GMm}{r^2}, \quad v_0^2 = \frac{GM}{r}$$

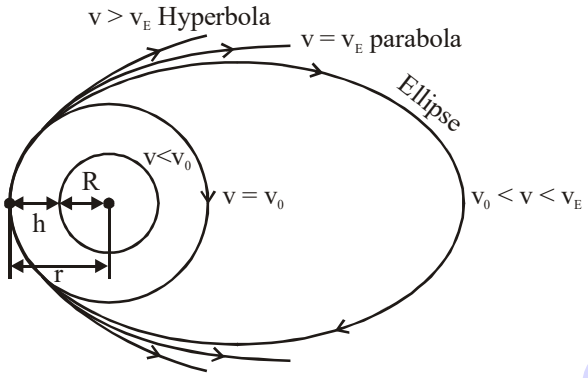
or $$v_0 = \sqrt{\frac{GM}{r}} = R\sqrt{\frac{g}{(R+h)}}$$

with $g = \frac{GM}{R^2}$ and r = (R + h)



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(b) Tangential velocity other than orbital velocity v_0 :



7. ENERGY OF SATELLITE

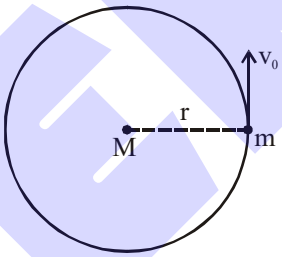
(a) In circular orbit

$$v_0 = \sqrt{\frac{GM}{r}}$$

$$KE = \frac{1}{2}mv_0^2 = \frac{GMm}{2r}$$

$$PE = \frac{-GMm}{r} = -2KE$$

$$TE = PE + KE = \frac{-GMm}{2r}$$



(b) In elliptical orbit :

$$TE = \frac{-GMm}{2a}$$

where a is the semi-major axis of the ellipse.

8. GEOSTATIONARY SATELLITE

(a) A satellite which appears to be stationary for a person on the surface of the earth is called geostationary satellite.

(b) It revolves in the equatorial plane from west to east with a time period of 24 hours.

(c) Its height from the surface of the earth is nearly 35600 km and radius of the circular orbit is nearly 42000 km.

(d) The orbital velocity of this satellite is nearly 3.08 km/sec

(e) The relative angular velocity of geostationary satellite with respect to earth is zero.

(f) The orbit of a geostationary satellite is called as **parking orbit**.

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